9 Video Games and Risks

Two Research Perspectives
The Active Media Perspective
The Active User Perspective
Other Questions
Final Remarks



The Walking Dead (2012), a zombie game

This chapter explores the risks involved with playing video games—or, in popular parlance, the potential harmful effects of video game play. This is an issue that everyone in the industry—and just as many people who have nothing to do with video games—seems to have an opinion about. For angry parents and determined teenagers, for dismissive developers and anxious educators, and seemingly for everyone in between, video games seem to contain an element of danger. Over the years, the question of harm has received massive attention in both public debate and research circles. Although the interest may not be as great as in the last millennium, the area has still regularly received attention, especially from the general media that often looks for causal links between shootings and playing video games.

Was the Columbine High School shooting of 1999 aided by the killers' fondness for violent video games? In 2011, did the Norwegian killer Anders Breivik plan and carry out his crimes with the help of computer games? Can violent video games make you more aggressive? Questions like these never seem far from the public agenda, and researchers of various stripes have for the last few decades tried to answer them.

We discuss this Pandora's box by contrasting two competing research perspectives: the "Active User" perspective and the "Active Media"

perspective. The discussion will make clear that the link between violent games and player aggression has received most attention but that beneath this issue lie even more basic disagreements about how we experience and perceive video games.

As we have seen throughout this book, academic research has been conducted on many elements of video games; however, the study of dangers in connection with games remains a key research avenue, as it continues to receive massive media attention and is still an arena for great controversy. As mentioned, research within this area has primarily been centered on the question of whether video games lead to increased aggression in players. This concern has been driven by regular bursts of public concern in relation to violent video games, most noticeable on the release of *Death Race* in 1976, *Mortal Kombat* in 1994, *Grand Theft Auto III* in 2001, *Medal of Honor* in 2010, *South Park: The Stick of Truth* in 2014, *Hatred* in 2015, and *Rape Day* in 2019.

Until the late 1990s, the majority of funded game research involved a risk perspective. This century, it has become somewhat of an orphan in games research, as more than a few researchers have flatly refused to take part in studies of risk. These researchers, as we'll discuss further, typically feel that video games as a medium are treated unfairly—that the discussion of games and aggression is essentially an attempt to turn games into a scapegoat for more complex societal problems; they also fear that the link to aggression is a precursor to censorship not leveled at other media, which is a valid concern given the continual attempts, especially in the United States, to pass laws to curb video games.[1,2] Also, Steam has over the years been under heavy fire to censor games on their distribution platform. Scholars who refuse to engage in this debate may well be interested in the aggression question itself but simply resent what they see as unfair criticism of video games, and suspect that opponents of games may have ulterior motives.

TWO RESEARCH PERSPECTIVES

Active Media perspective: A school of thought that believes that media actively influence a mostly passive recipient, the player.

The **Active Media** perspective refers to a school of thought that believes that media actively influence a mostly passive recipient, the player. Its proponents typically use "classical" methods of research, often based in a laboratory, and are influenced mainly by social psychology and behaviorism. This perspective mostly exists outside mainstream games research; "risk" researchers do not consider themselves games researchers, do not attend games conferences, and often do not play games. This split has at times led to problematic research, lacking a basic knowledge of (or interest in) video games, although recent years have seen more rigorous studies.

Active User perspective: A school of thought that stresses the active interpretation and filtering players exhibit when playing video games.

In contrast, the **Active User** perspective refers to a school of thought that stresses the active interpretation and filtering players exhibit when playing video games. The Active User perspective will gather empirical data from natural settings and will tend to look to anthropology, cultural studies, and media theory for theoretical ammunition.

We should note that the distinction between the Active Media and Active User perspectives is an artificial one, set up by us to recognize some basic differences between researchers examining the risks of computer games. In the following pages the Active Media perspective and the Active User perspective are discussed in a way that provides the necessary knowledge for understanding the debate on risks, both in research circles and in the general press. The chapter should also give the reader a starting point for conducting studies in both areas with an eye to potential problems and limitations.

Over the last 35 years, more than 200 studies have been conducted to try to understand the relationship between video games and risk. Taken as a whole, this body of knowledge has been influenced more by the Active Media than the Active User perspective. The former has imbued this research with a social psychological perspective, and a reliance on what Active Media researchers refer to as "the effect tradition," that is, an almost automatic relationship between playing violent video games and becoming more violent. In the Active Media perspective, video games are conceived as having a direct, objective, and measurable effect on players.

An Active User perspective, which takes the opposite starting point, has over the years tried to challenge this tradition. According to researchers from this latter perspective, video games do not have the same effect on everybody but are mediated by a variety of factors such as playing context, genre expectations, and the individual player's interpretation. The player is not seen as a passive recipient of the content in video games but rather as an active, selective, and critical user.

In the sections that follow we present typical research designs for each perspective. It should be stressed that in principle the two perspectives (or "theories") are not directly linked to specific academic fields, although researchers within each are typically allied with certain methodologies, as we'll see below. These research designs summarize the specific ways that each group prefers to conduct a scientific study. By examining a standard example of these two research approaches, we hope to show that the way the question of harmful video games is formulated and examined will to a large extent decide the answers arrived at.

A Typical Research Design for the Active Media Perspective

A typical study in this tradition will start out with two groups of subjects; they can number anywhere from 8 to 100. One group plays a violent video game (for instance, the action game *Quake*) while the other group plays a nonviolent video game (for instance, the racing game *Need for Speed*). Before subjects start playing, their level of aggression is measured—different methods may be used, from a standard multiple-choice psychological questionnaire to a physiological test that involves taking swabs of saliva to find the amount of cortisol in the player's body. This measurement is repeated for both groups after they have played their games. The assumption is that any difference in

the change in aggression levels between the two groups can be attributed to the difference between the two games. The effect of one violent game can then be generalized to other violent video games that these groups might play.

A Typical Research Design for the Active User Perspective

In a classic study following an Active User perspective, the researchers observe and interview a group of children playing different video games in a natural setting (such as an after-school club) over a relatively long period (say, three months). Through their observations the researchers note behavioral characteristics such as verbal and physical violence. Questions could range from the children's perception of dying in a game to whether they have bad dreams. The researchers aim to describe the full experience of gameplay and the context in which these children play games; the ultimate goal is to understand how the players integrate video games into their daily life—how players use the games.

The effect of playing certain games (or types of games) is just one potential focus of an Active User study and is rarely looked for specifically. When incidences of aggression are found, they are analyzed in the specific context in which they arose; researchers hesitate to generalize actions or behaviors to other situations because they believe that each player perceives each game differently and the context of play is important for the player's perception. Furthermore, because everything that the children do is seen as dependent on the context (in this case the after-school program, the particular group of subjects, and the specific games they play), the researchers are not able to isolate "violence" as a variable.

The focus of these two research designs makes all the difference. The Active Media perspective looks at what a video game does to a player, whereas the Active User perspective is interested in what the players do with the video game. This crucial difference stems from a very basic theoretical disagreement on how we should understand people's psychological functioning. This split also runs through a number of other disciplines and may be exemplified by the disagreement evident in the statements made by the two groups.

An Active Media proponent concludes, based on laboratory studies, that

> exposure to violent video games is significantly linked to increases in aggressive behaviour, aggressive cognition, aggressive affect, and cardiovascular arousal, and to decreases in helping behaviour.[3]

Compare the following statement from a group of Active User proponents:

> Many scholars believe that trying to understand the media's impact on human development through laboratory measurements and other numerical methods is inherently flawed.[4]

Table 9.1 The differences between the Active Media perspective and the Active User perspective

	Active Media perspective	*Active User perspective*
Scientific and theoretical starting point	Behaviorism, social psychology, experimental psychology	Anthropology, ethnography, literary studies/ semiotics, cultural studies, media theory
Research methods	Predominantly quantitative (controlled studies)	Predominantly qualitative (interview and observation)
Main interest	The effects of media on attitude, perception and behavior	Meaning, role, and function of media
Research object	The video game (what effect the game has on the player)	The player (what the player does with the game)

Though Table 9.1 simplifies each perspective slightly, laying out their basic characteristics side by side gives us a graphic appreciation of the differences. Whereas the Active Media perspective involves generalizing, researchers within the Active User perspective usually avoid making general claims. The Active Media perspective examines the specific influence of violent video games, while the Active User perspective examines video games in a broader way. Researchers within the two perspectives do share some notions of what qualifies as science—both groups agree, for example, that logic, argument, and reasoning are pillars of science. However, it is unlikely that studies taking an Active Media perspective will ever fully convince Active User researchers and vice versa. As we look at the basic assumptions of the two groups, it seems less likely that they are directly disagreeing with another and more likely that they are having different conversations. Their theoretical assumptions are dramatically different from each other.

Attempts have recently been made to combine the approaches to achieve a stronger research design and to facilitate better dialogue between the two perspectives.[5] Traditionally, the two research positions have seldom engaged in direct discussion of each other's research; the occasional public debates are centered on the Active User perspective's skepticism toward the basic premises of the Active Media perspective. Within the last five years, however, the Active Media perspective has begun to engage with the criticism, trying to refute some of the critiques.[6] Media researcher David Buckingham criticizes both positions for building barriers but also states that "despite the more complex views of meaning which have been developed, much of the work on violence has remained stubbornly tied to behaviorist assumptions."[7]

Let us deepen our understanding of the risk issue by exploring the thinking behind each of these perspectives, as well as the relevant criticisms of each. We will begin with the Active Media perspective, since it remains the most dominant perspective.

THE ACTIVE MEDIA PERSPECTIVE

The Active Media perspective has set the agenda for research into the risks of video games since the early 1980s; though it is less dominant today, its influence continues to be felt, with more than 130 published studies in the last two decades. The Active Media perspective has its theoretical roots in North American psychology, and the "behaviorist tradition" constitutes an important part of its foundation. Another basic building block is communication theory (often defined as "who says what to whom in what channel with what effect"), particularly the branches inspired more by classical social sciences than by hermeneutic approaches to media use.

With its origins in the 1910s, behaviorism conceptualizes human beings as fairly simple systems that respond to the stimuli around them; classical behaviorists avoided discussing inner mental processes. The Active Media perspective relies on behaviorism, especially its use of theory for understanding empirical results but is offering still more sophisticated theoretical frameworks to guide predictions and interpretations.

Active Media research can technically employ most types of methodologies, from experimental studies to cross-section correlation. Nevertheless, these studies share a basic assumption that a particular medium (such as video games or violent video games) has one particular effect on people (who are basically similar); this assumption often leads to experiments with a fairly standard research design, similar to the example presented earlier.

The most used research design is usually referred to as an experimental study. Ideally these take place in a laboratory setting, and by filtering out confounding variables the researcher aims to measure the exact effect of a given variable. The two other typical research designs are cross-section correlation studies and longitudinal correlation studies.

Cross-section correlation studies look at a group's use of video games and level of aggression at a given point in time. A correlation is said to exist if the players of violent video games are more aggressive than the nonplayers. So, if a group of gamers plays more violent video games and is more aggressive than others, this denotes a correlation.

Longitudinal correlation studies typically follow a given group of people over a number of years. If, over time, a correlation is ascertained between the fact that those who play violent video games also exhibit more aggressive behavior (and that behavior was not present at the start) then there is a causal correlation: violent video games in those cases lead to aggressive behavior. There are still few studies of this type that have been undertaken and none really before 2005, even though they potentially offer the strongest evidence for establishing a clear cause-and-effect relationship between violent video games and aggressive behavior.

Following a lot of criticism, research designs have been improved and attempts have been made to review earlier research in the related field of meta-studies (which we'll discuss later). There is still disagreement among researchers within the field that the results from these three typical effect

studies are not strong enough by themselves to give a final verdict on whether video games promote aggression. However, overall, Active Media researchers are increasingly convinced that video games pose a threat that needs to be dealt with.

We will return to some concrete research results after we have looked at common theories in the Active Media perspective.

Theories and Methods

Researchers within the Active Media perspective have a variety of theoretical starting points. Albert Bandura's social learning theory from the 1970s has often been preferred, but over the years other approaches have been applied. Here, we will introduce the ones that have historically played a role: catharsis theory, cultivation theory, social learning theory, general arousal theory, the cognitive neo-association model of aggression, and the general aggression model. See Dill and Dill,[8] Calvert,[9] and Barlett and Anderson[10] for other introductions.

- *Catharsis theory*:[11] The idea of catharsis has a long history in psychology, dating back to Sigmund Freud in the late 1890s, but in video games research a newer version of the theory is often used. Seymour Feshbach and Robert Singer proposed this theory in the early 1970s, pointing out that experiencing depictions of violence in media (whether video games, movies, or television) can actually reduce aggressive feelings. The idea is that one's own aggressive feelings will be mirrored in the media, and experiencing them will reduce internal tension. In video games, the active role of the player further enhances this effect as the aggressive feelings are not only seen on the screen but also actively performed by moving an avatar on the screen.

  The application of catharsis theory within the Active Media tradition is limited to a few studies, but the theory is often used in less scientific discussions. The limited interest in catharsis as an explanatory theory could also be attributed to the inclination among effect researchers to focus on the dangers of video games; they are less likely to seek out arguments that games reduce aggression. In the last decade, the catharsis theory has also become less acceptable in psychology in general.[12]
- *Cultivation theory*:[13] Cultivation theory is built on television research, and attempts to explain to what degree media lead to a distorted perception of social reality, for example, through stereotyped perceptions. The theory has been less used in relation to video games; though there are certainly characters in video games that are based on ethnic or gender stereotypes, it is still uncertain what stereotyping, if any, video games can produce in the player.
- *Social learning theory*:[14] According to social learning theory, behavior is learned through imitation of attractive models with attached rewards. Supporters stress that video games exemplify this model of learning: they

demand the player's full attention and entail active identification with characters on the computer screen. Furthermore, rewards are attached directly to the performance of symbolic violence (getting points, for example, by killing opponents). Therefore it is more likely that these aggressive actions are transferred to the outside world.

- *The general arousal theory*:[15] This theory claims that video games will increase the player's arousal level and thus increase his energy and the intensity of his actions. The increased arousal will not necessarily lead to different player actions but more likely a heightened intensity in these actions. Supporters of this theory point out that especially violent video games provide the arousal necessary to facilitate more aggressive behavior. Getting into an argument with a friend while playing a violent video game may accelerate into a fierce discussion, or you may be more likely to get irritated when someone pokes you after playing a game than after talking on the phone or washing dishes.
- *Cognitive neo-association model of aggression*:[16] This theory holds that violent media lead to hostility and aggression due to reinforcement of related association nodes in the brain. These nodes can be created and strengthened by video games; through them, aggressive thoughts will be transformed into physical action.
- *General aggression model*:[17] The general aggression model (GAM) is one of the newest theories in the field and combines earlier theories on aggression with cognitive schemata theory. Today this is the de facto standard in the field, and there has been made extension of it over the years. Cognitive schemata theory suggests that we order our experiences in different schemata which we use when we perceive new experiences. According to the model, violent media content causes aggressive behavior by influencing the person's internal state, which researchers measure by looking at a variety of cognitive, affective, and arousal variables. Violent media content increases aggressive behavior in several concrete ways: by teaching the player how to perform aggressive actions, by influencing underlying aggressive and cognitive schematas, by increasing arousal, and by creating an emotional and aggressive mental state.

From these descriptions it should be clear that most theories refer to media in general; the one exception is the general aggression model, which was developed with video games in mind. The theories, mostly developed in the 1970s and 1980s, reflect the strong influence of behaviorism and cognitivism at that time. This is also true of most of the research questions posed, research designs constructed, and methods used. They are also heavily inspired by research on television and violence.

Important Studies

In the last 15 years, Active Media research has to a large extent been pushed forward by meta-analyses of earlier studies and the emergence of

longitudinal studies—in other words, looking at previous studies and combining their results to make a stronger case for the risks associated with video games, while looking at the longer term effects. However, there are some notable exceptions that we will cover here. As noted earlier, the total number of studies is at least 130 but is hard to estimate, and depends on what one chooses to include as a research study.

In their 2000 study, Anderson and Dill examine video games in relation to aggressive thoughts, feelings, and behavior, both in the artificial setting of a laboratory and from data on real-life behavior. This study consists of two parts; the first part is the most interesting, as it represents a classic research design and findings typical of the effect perspective.

More than 220 psychology students participated in the first part of the study: a majority were women, the average age was 19, and the students were awarded course points for participating. The subjects filled out a questionnaire describing their five favorite games, and their experience with the content, violence, and playing style of these games. The students were also asked about the amount of time they spent playing. From these questionnaires the researchers extracted information about the subjects' aggressive behavior, criminal record, game habits, and general worldview.

The results showed that male players felt more secure, played more violent video games, and spent more time playing than women. The more a participant played violent games, the more likely it was that he or she had engaged in criminal activities. Academic achievements were also hampered by a high playing time. Those who played violent video games did even worse academically, and were more involved in aggressive crimes like theft.

The second part of the research consisted of an experimental study. It confirmed that violent video games reinforce aggressive thoughts, with men being more aggressive than women. The overall conclusion was that there is a causal relationship between realistic violent video games, aggressive behavior, and crime.

In their 2002 study, Active Media researchers Steve Durkin and Bonnie Barber examined a number of links between video games and adolescents. There were 1,304 participants, all adolescents, covering a broad socio-demographic profile and with an average age of 16. The study captured both positive and negative effects of video games. The researchers divided the participants into three groups based on their video game usage and rated them according to the following variables: family closeness, activity involvement, positive school engagement, positive mental health, substance use, self-concept, friendship network, and disobedience. The overall conclusion was that "computer games can be a positive feature of a healthy adolescence."[18] Though noting the potential negative impact of video games, Durkin and Barber also found that games can offer a variety of positive effects in the lives of adolescents; the optimism of their work is in surprising contrast to traditional studies within the Active Media perspective.

The study measured the game use of youngsters on a scale from 1 to 7 (1 = never, 7 = daily). Looking more closely at the results, it is clear that

the group with low usage of video games (defined as answering 2, 3, or 4) rates better than both the group that does not play video games and the group that plays video games regularly (defined as answering 5, 6, or 7). The results indicate that the group with low usage has less depressed mood, better self-esteem, and better academic achievement. The group that does not play video games at all is more deviant and is more prone to criminal behavior. The study did not find differences between the groups in relation to self-reported aggression.

A 2001 study by Thomas N. Robinson, Marta L. Wilde, Lisa C. Navracruz, K. Farish Haydel, and Ann Varady stands out for its nontraditional research design: it looks at what happens when we change children's media habits. The study involved children from two different schools with an average age of 9. The goal was to reduce media consumption (television, video, and video games) and examine the consequences. The participants were split into two groups: the first group maintained its normal level of media usage, without outside interference. The second group was allowed to maintain its normal media habits but was educated about media use during a six-month course. The course used social learning theory to help the children explain what happened to their behavior and emotions when they used media. It also introduced turn-off television days, a parents' information campaign, and discussions in general. After the course, the children were encouraged to reduce their media consumption to around seven hours a week.

The results showed that the trained group was perceived as less aggressive by their friends and showed less verbal aggression. Researchers also witnessed a tendency toward less physical aggression, but the decrease was not statistically significant. This study is particularly interesting because it points to realistic ways to limit the influence of aggression by educating players of video games to become critical and self-aware. This is quite different from the traditional Active Media perspective, where players are seen as helpless victims in the hands of video games. It also turns the research design upside down, in comparison with traditional Active Media research. Instead of measuring what happens when one plays video games, it measures what happens when one does not. Of course the problem with the above study is that it doesn't focus exclusively on video games but media use in general.

Another recent trend in examining the effect of violent video games is to apply more advanced statistical analysis than previous studies. This has revealed interesting problems with earlier research. Earlier studies had looked primarily at whether there was any relation between playing violent video games and measures of aggression—not at how big the effect of playing violent video games was. These early studies found that, yes, you become more violent when playing video games. However, they did not say anything about the size of the effect of video games compared to other factors such as parental involvement. Attempts to remedy this problem are seen in more recent studies from 2004, for example, by Gentile and colleagues[19] and by Funk and colleagues.[20]

Studies carried out in the last ten years have increasingly attempted to introduce more subtlety into their analyses. In 2004, Douglas Gentile and colleagues found that adolescents who played video games with more violent content were more hostile, got into more arguments with teachers, and were more involved in physical fights compared to their contemporaries who didn't play violent video games to the same extent. These adolescents also seemed to perform worse in school. The superficial evidence pointed to a damning correlation between video game violence and a broad spectrum of problematic behavior.

However, when the researchers included additional variables—especially the gender of the subjects and involvement of their parents—the results were radically different. When these factors were considered the effect of violent video game exposures on physical fights practically disappeared. When you look only at the relationship between violent video games and violence, underlying variables may actually account for the apparent link. For example, let's say that less parental involvement leads to more playing of violent video games. Let's also say that less parental involvement leads to more violence. From a simple statistical analysis you might conclude that violent video games lead to more violence. However, this would not be a valid conclusion, as in this example the true explanatory factor is parental involvement.

These findings stress the importance of looking for additional variables, rather than limiting studies to focus exclusively on the relation between playing violent video games and aggression. Ultimately, the study indicated that parental involvement and gender differences were a more powerful underlying explanation for aggression than video games. This lends some support to Active User criticism of the Active Media perspective for using overly simple models of humans and their behaviors.

Attention deficit/hyperactivity disorder (ADHD) is a relatively newly defined and controversial condition, which some have associated with video games. A 2010 study links exposure to video games with ADHD. In the study, 1,533 participants with ages ranging from middle childhood to early adulthood were assessed during the course of a year.[21] For younger participants, parents, teachers, and the children themselves reported on media exposure and attention problems, while older participants used self-reporting. The study found that, for all ages, exposure to video games (and television) was correlated with ADHD. It was suggested that video games may be a predictor. It also seems that longer exposure to video games increases the risk, likewise that the risk is higher the younger you are, and higher among males. The authors warn that the nature of the experimental setup does not make it possible to account for any unknown factors that might explain some of the correlation. Finally, it is stressed that the catch-all of "video games" is problematic, as there is a huge difference among different types of video games.

Longitudinal Studies

Active Media perspective research is often criticized heavily for looking only at short-term effects. One 2005 study, however, examined the effect

of violent games one month after play. The study was conducted with 213 participants aged 14 to 68 (with an average age of 28), who played *Asheron's Call 2* for one month. The study found no increased aggression on any measure after the trial period.[22]

More rigorous studies have followed, however: in 2008 a joint US-Japanese study included 1,595 subjects with an age range of 9 to 18 years, who were measured on video game use and physical aggression at the start and end of a school year. The study found that those who played a lot of violent video games became relatively more physically aggressive across genders. It also found the effect of video games to be greater among younger students. Finally, it demonstrated that the effect of violent video games on physical aggression was not mediated by living in a collective culture (Japanese) or in a more individualistic culture (United States).[23]

Krahe and Möller's 2010 study followed 1,237 German high school students over 12 months through self-reporting on media use including video games. It found that media violence is correlated with physical aggression and lower empathy, albeit with a quite small effect size of 2 percent—meaning that only 2 percent of the variation in aggression can be explained by the differences in media usage. This study did not focus exclusively on video games but does offer some valuable longitudinal evidence.

In a study from 2011 by Ferguson, Miguel, Garza, and Jerabeck, 165 youngsters were followed to examine whether youth aggression could be a consequence of playing violent video games. The study had an initial interview, a one-year follow-up, and a three-year follow-up. The study found no relation or negative outcomes based on exposure to violent video games. Rather, family violence and peer influence was found to be the biggest influence.

A later study from 2012 by Willoughby, Adachi, and Good surveyed 1,492 youngsters in grades 9 through 12 about their video game habits and aggressive behaviors. Interestingly, the study looked at the classic issue, which is the selection versus socialization hypothesis. The selection hypothesis implies that a person selects aggressive video games because they already have a preference for aggression, whereas the socialization hypothesis predicts the subject becomes aggressive from playing aggressive video games. The study supported the socialization hypothesis: It was found that violent video games predicted higher aggression, whereas nonviolent video games did not.

In a German study from 2014 by Mößle, Kliem, and Rehbein the effect of violent video games on aggression was similarly examined. Special focus was given to the role of empathy as a moderator of the effect of violent video games. The 1,207 school children in the study were on average 10.4 years old at the start of the study and 12.4 years old at the end. The results showed that there was a relation between violent video games and aggression. However, it was also found that empathy was a key variable to mediate the effect of violent video games, although only in boys. This may suggest that violent video games only have a negative effect when empathy is not developed

through other appropriate interpersonal activities. Thus, it may not be so much the exact effect of violent video games that is crucial to examine but rather whether players have developed important empathy skills outside the game world.

In 2018 a new longitudinal study supported a cautious approach to blaming video games for aggression. In the game the researchers made three different interventions over a two-month period (*GTA V, The Sims 3*, and no game) with 90 participants aged 28 years with even distribution of gender. The participants needed to play the game at least 30 minutes a day during the study. To measure aggression, several standard questionnaires were used but none of them found any effect after the two months.[24]

The efforts to improve the rigor with which violent video games is examined reached a new high in 2019 with a study focused on providing a better research design, and especially changing the method to measure aggression. The study examined 1,000 British adolescents aged 14–15 years. The study asked participants for the top three games they played, and then coded them based on official game ratings to see if they were associated with higher report aggressive behavior or less prosocial behavior. The aggression was measured by asking parents to report real aggressive behavior. The study found no link.[25]

This is in line with past criticism of researchers that even if you can measure indirect effects of aggressions in children after playing violent video games this does not translate into real-life behavior, a point that was again stressed by Fergusson along with colleagues in Society for Media Psychology and Technology under APA.[26] Indeed, nobody has so far been able to find relationship between video games and real-life violence on a societal level.[27]

Meta-Studies

A large number of meta-studies (studies combining the results of previous Active Media research) have been conducted in the past 15 years. Though they all seek a larger sample, and thereby a stronger dataset for making statistical analyses, these studies vary in their intentions. Some simply review earlier studies and look for overall tendencies. The more ambitious collate original data from earlier research and carry out new statistical analyses. Here we present examples of both types, as they have different strengths. The conclusions from the newest of these studies, conducted within the last 15 years, have become increasingly mixed. The main differences among conclusions of the meta-studies is how seriously they take the problems of method presented in most earlier studies. Does one dismiss these studies entirely because they are flawed, or does one insist that the relationship between violent video games and aggression (that many studies find, although not all) should lead to a cautious approach until stronger method designs are available?[28]

In their 2001 meta-study, psychology professors Craig Anderson and Brad Bushman offer a straightforward and potent conclusion: "Violent

video games increase aggressive behavior in children and young adults."[29] This conclusion is strengthened by the study's critique of the flawed methodology of many previous studies. The authors focus on the difference in effect size between flawed experiments and stronger experiments, and find that the effect size (the actual harmful impact of video games, not just the strength of the relationship) increases with the strength of a study's methodology. The conclusion is that the current combination of correlation studies and experimental studies provides strong evidence that violent video games lead to aggressive behavior in real life.

Communication researcher John Sherry does not disagree, although he is less convinced. He stresses that the effect of video games is smaller than that of violent television.[30] British psychology professor Mark Griffiths is even less convinced, stating that the only consistent finding from the many studies he has analyzed is that young children are affected by violence more than teenagers and older children.[31]

Psychologist Jonathan Freedman stresses that the research is limited and finds it very problematic that strong claims are made. He states:

> I cannot think of another important issue for which scientists have been willing to reach conclusions on such a small body of research. Even if the research had been designed and conducted perfectly, there is far too little evidence to reach any firm conclusions.[32]

As the meta-analysis has gained in strength, it has attracted its own criticism, including publication bias and unreliable statistical measures.[33] Some see the meta-analysis as the ultimate answer to the question of video games' malign effect. However, one should remember that all the criticism of methodology and theoretical assumptions still holds true, no matter how many such studies are carried out. Simply put, 300 × 0 is still 0. This is not to say that the Active Media approach is not making headway, but the meta-analysis should be approached with open eyes, particularly in view of the complexity of these studies, which has led to controversy. This is very clearly stated in the following quote:

> Meta-analyses of video games similarly produce weak and inconsistent results. Two early meta-analyses claimed to find small but significant effects for video game violence on aggression, although a subsequent review of these meta-analyses during a court case revealed that the authors may have simply ignored research that didn't fit with their hypotheses (ESA, VSDA and IRMA v. Blagojevich, Madigan, & Devine, 2005).[34]

In their meta-analysis from 2010, Craig and colleagues include 136 studies on aggressive behavior, including most studies within the Active Media perspective. The study does respond to a number of the criticisms leveled at previous meta-studies. However, as many of the studies are duplicates from

previous meta-studies, some problematic elements have been carried over. The authors conclude:

> we believe that debates can and should finally move beyond the simple question of whether violent video game play is a causal risk factor for aggressive behavior; the scientific literature has effectively and clearly shown the answer to be "yes." Instead, we believe the public policy debate should move to questions concerning how best to deal with this risk factor. Public education about this risk factor—and about how parents, schools, and society at large can deal with it—could be very useful.[35]

Despite the strong stance of Anderson and Bushman, a large number of Active Media researchers favor Freedman's more cautious approach.[36] The exceptions are usually in research studies either co-authored by Craig Anderson or by his colleagues.[37]

In recent years, Christopher Ferguson has become one of the most vocal critics of the casual link between games and violence. In early 2015, he released a study that changed the battlefield by looking at the actual societal impact of more media violence. Previously, this has often been anecdotal evidence that would go something like this: If violent video games were so bad wouldn't our society have collapsed with the rapid rise of violent television and video games during the last 20 years? In this approach, Ferguson looks at how much violent video games were played during specific years (1996–2011) and how this correlates with violent crimes in the same period. Indeed, he finds no effect—actually he finds an inverse effect, which would suggest that the opposite was true. Although he makes no such claims, it suggests that the findings are quite strong.

However, the methodical weaknesses of such a study are legion. There are many potential confounding variables when you look on a societal level, all of which need to be controlled for—for example, the general development in equality and income level in society at large. Also, how do you measure what people actually play? Ferguson uses the most sold games from the Internet Movie Database (IMDb), but sales data has become quite weak since the mid-2000s with the emergence of online and mobile games, where data is fragmented.

The most recent meta-analysis of the relationship between violent video game play and physical aggression over time came out in 2018. Here Prescott, Sargent, and Hull look at 24 studies with a total of 17,000 participants. The study does find a link strongest with white participants, less strong among Asian participants, and not reliably for Hispanic participants. Despite a small effect size that explains only around 1 percent of the variance, they stress that this is still practically significant. As APA attests to whether an effect size is large or small in a given domain depends on the eye looking—in other words, this is a subjective question.[38]

It is also noteworthy that a number of healthcare associations have made strong statements about the risks of video games—for example, the

American Psychiatric Association has spoken out against them—although here there is also disagreement.[39] In 2007 the American Psychiatric Association refused to include video game addiction in their diagnostic system, known as the DSM-IV, due to limited and inconclusive research: "the APA does not consider 'video game addiction' to be a mental disorder at this time."[40] However, with the DSM-5 in 2013, "Internet Gaming Disorder" was included on the watch list warranting more research, and it is likely that APA will update its stance on the issue in the coming years. In 2015 the APA Task Force on Violent Media published a new report to bring APA's position from 2005 up to date with most recent research results, and revise their resolution on violent video games.[41] Here, they concluded:

> "The research demonstrates a consistent relation between violent video game use and increases in aggressive behavior, aggressive cognitions and aggressive affect, and decreases in prosocial behavior, empathy and sensitivity to aggression. . . . No single risk factor consistently leads a person to act aggressively or violently," the report states. "Rather, it is the accumulation of risk factors that tends to lead to aggressive or violent behavior. The research reviewed here demonstrates that violent video game use is one such risk factor."[42]

Based on the technical report the APA put into place a resolution with a number of warnings in relation to video games.[43]

Overall, meta-studies do find a relation between violent video games and aggression albeit the discussion remains what the small effect size shows—and that no real link has yet been established on a societal level. Ultimately, the results are too mixed to warrant any strong conclusions, although the group of researchers around Craig Anderson are very firm in concluding that video games lead to aggression. Even if there is a correlation, and some effect size, it seems that the link to actually observable aggression from playing video games is weak.

Indeed, numerous researchers are far from convinced, and in line with common research practice it would be premature to conclude that there is a link between violent video games and real-life aggression. However, there are enough indications to warrant further studies, and a cautious approach. And those who prefer can conclude from these studies that a cautious stand is the safest: they can avoid violent video games, and in doing so avoid any potential (though not yet proven) long-term real-life harmful effects.

Criticism

Criticism of the effect perspective has grown with surprising intensity over the last 15 years, and has been more explicit than that leveled at the Active User perspective. The criticism has been focused at different levels, spanning basic differences over ontological approach, disagreement about what constitutes scientific evidence, and specific faults in methodology. Criticism

directed at the competing Active User perspective has also grown stronger over the last ten years, however. Psychology in general has broadened its approaches and challenged basic assumptions about how to do research and what methods to use. Some are inclined to disqualify all published Active Media studies due to severe problems of method. The influential comparative media scholar, Henry Jenkins, is very skeptical of the simplistic conclusions often put forward. He cautions against looking for links in studies that do not reflect real-life media use.[44]

The primary criticisms leveled against the Active Media perspective are described in the sections that follow.

Laboratory Versus Everyday Situations

The Active Media perspective relies heavily on laboratory studies to establish a relationship between video games and aggression. However, the laboratory is not similar to everyday situations, and although the differences may seem obvious, Active Media researchers continue to apply results gained in an experimental setting to real-life behavior. The majority of laboratory studies do a very bad job at replicating the phenomena they want to study. Important variables—such as the social experience around the video game, the player's control of the situation, and the desire to play video games—are absent. Usually, players are not in charge of how long they play, what they play, or how they play. This is hardly representative of an activity that is mostly driven by pleasure.

Another distortion created by the laboratory is that the length of the play session in these studies can vary from 4 to 75 minutes. This range could very well have an important bearing on how the video game is experienced. The players with brief play periods are abruptly interrupted, and although this may also happen when you play normally, the short time span is not representative of most play. In any case, the interruption is arbitrary, and not linked to the actual experience of playing violent games. Participants given a long play duration may also experience something different from real life, where they may be used to taking turns or playing for shorter periods. The longer play time may also, especially for inexperienced players, lead to tiredness. This tiredness then becomes a confounding variable, which can distort the results. Sherry's 2001 meta-analysis finds that less aggressive behavior is exhibited the longer one plays. This seems counterintuitive, and could unintentionally demonstrate that the artificial setting of the laboratory may lead to an increase in aggressive behavior. In long play sessions, on the other hand, players may be less aggressive because they get used to the artificial situation and hence do not feel uncomfortable.

Within the structures of the laboratory, participants are inevitably led in specific directions by a study; there is always the danger that they might anticipate what the researchers are looking for and adjust their answers accordingly. Though we can't prove a connection, look at the different results from the studies we examined earlier by Durkin and Barber[45] and Anderson

and Dill.[46] Each research team began with different expectations: how can we be sure that these expectations didn't influence the different results they obtained? Due to their starting point the majority of Active Media researchers expect to find a negative effect; not surprisingly, the majority of them do.

Problems of Causality

A second source of criticism is the problem of causality, which Ferguson[47] also refers to as the "third variable" effect. Even within the carefully controlled laboratory environment, we cannot be certain how to explain a given effect. If players of violent games exhibit aggressive behavior, how do we know this isn't because they weren't able to choose the game themselves, and were "forced" to play a title they didn't like? Such problems are even more pronounced in cross-section correlation studies. Here, a study may find that people who show a preference for violent games exhibit more aggressive behavior than those who play more peaceful games, or who do not play at all. But from this result alone causality cannot be established: aggression could be caused by the violent games or the preference for violent games could be caused by the person's aggressive tendencies. In the case of longitudinal correlation, this problem is less acute. However, such studies cannot completely assure validity. The gaming variable may theoretically be tied to other variables—anything from parental involvement to gender, as we saw earlier—which may in fact be the actual cause of the increased aggression. Theoretically, we can even go so far as to say that the aggression could be caused by the social stigma assigned to players of violent games, rather than by the games themselves.

Defining Aggression

Another basic problem in studies within the Active Media perspective is the inconsistent use of the term "aggression": both what constitutes aggression and how it is measured. This problem is not limited to media studies but applies to studies of aggression in general.[48] Researchers have used all of the following as measures of aggression: aggressive thoughts, hitting a doll, playing patterns, verbal aggression, aggressive behavior, physical fights, or the more obscure willingness to donate money to charity. These different variables in turn draw on different theoretical traditions that can be contradictory in their assumptions and unclear in their measurement criteria.

It is interesting to note that Geen[49] offers a very simple working definition of aggression: actions that do harm or the intention to do harm. This disqualifies a number of studies into the effects of video games, which instead look for violent thoughts or preferences for certain aggressive words. However, Geen acknowledges that underlying variables, such as those presented by Anderson and Bushman in their general aggression model, can mediate aggression. Aggression continues to be an ambiguous concept, and the exact relation between aggressive schemata, aggressive play, verbal aggression, and physical aggression is unresolved.

Video Games Are a Rich and Varied Phenomenon

The Active Media perspective does not approach video games as a varied phenomenon but implies that they are homogeneous, with only slight differences in their content. For most researchers within the Active Media perspective, the only relevant difference seems to be whether this content is violent or nonviolent. This approach would be problematic even with other media. To compare the effect of a nonviolent soap opera with a violent action movie, for example, would hardly make sense. They use very different artistic effects, and aspire to different goals, and one runs the risk of not measuring violent content but simply the general impact of different formats.

The enormous differences between video games are obvious to video game researchers but seem less clear to psychologists and other Active Media researchers from other traditions. Despite the evolution of game research over the last 20 years, a certain degree of ignorance remains about even the most basic differences between video games. A somewhat dated study by Anderson and Dill[50] loses its persuasiveness when the researchers choose to compare *Wolfenstein 3D* (a violent action game) with *Myst* (a peaceful adventure game). When we recall the large differences between the adventure genre and the action genre, we cannot help but doubt the study's implication that the only important difference between the games is the level of violence.

Statistical Problems

Ferguson, in particular, has in his research been critical of the statistical implications of the Active Media approach. In his 2007 study he shows that "studies employing less standardized and reliable measures of aggression tended to produce larger effect sizes."[51] He has also documented a tendency for studies showing greater effects of videos games on aggression to get published more frequently. In two other articles, Ferguson points out that the standards accepted for evidence are quite low, compared to other research areas, and that the so-called clinical cutoff point is not really applied. The clinical cutoff is the point at which an observed behavior becomes pathological. Hence, it may be that someone consistently becomes more aggressive from playing video games, but that doesn't mean that they become aggressive enough for this to actually lead to malign or pathological behavior. This is supported by the lack of any increase in violence on a societal level at a time when the popularity of games has increased massively.

Criticism of Basic Assumptions

Throughout this chapter, we have argued that the starting point for the Active Media perspective in examining the effect of video games is problematic. When investigating the relation between video games and aggression, opponents of Active Media have argued that a researcher ought to look in the places where aggression is actually found. They might, as a starting

point, interview hardened young criminals, look at whether video games trigger fights in clubs or bars, or visit the local arcade. Beginning with actual events (rather than in a laboratory) results in a hypothesis informed by empirical data. The researcher can then analyze this data and devise further tests to see if the results can be generalized to other real-life situations.

Strangely, Active Media researchers do not examine the actual places where a direct link between video games and violent behavior might be found. It is interesting, as Jenkins points out, that in the few studies of arcades, we find very little violence in the places where games are actually played.[52] We still need to establish a real-world link between playing video games and violence, beyond the often overhyped media descriptions of single, tragic events like the Columbine shootings, for which, on closer examination, the link to video games is at best sketchy.

Professor David Gauntlett has also suggested that the Active Media perspective offers a conservative perception of children and young people, one that runs contrary to modern developmental psychology. In modern psychology, as mentioned earlier, context is essential for framing an experience; individuals, far from being passive, construct their understanding of a situation. Recognition of humans' basic ability to interpret experiences is lacking in Active Media research. For Active Media researchers it seems irrelevant whether a player is killing someone to save the world or killing someone for the fun of it.[53]

Ultimately, one may have to accept the fundamental assumptions of positivism and behaviorism if the Active Media perspective is to be meaningful. But in the United States and much of Europe, as well as many other places around the world, these theoretical assumptions are no longer the dominant psychological approach. The Active Media perspective remains haunted by its inherent problems of method, mentioned above, and by a sustained and withering critique of its basic theoretical assumptions. Though its legacy remains strong, these problems have limited the impact that Active Media research has had on the broader video game research community.

THE ACTIVE USER PERSPECTIVE

If we can generalize, the video game research community typically approaches its work from an angle closer to this perspective. Active User researchers are mostly from the humanities, particularly from the fields of cultural studies, anthropology, and pedagogy; these researchers share many of the same theoretical assumptions about people, methods, and what research entails. When dealing with video games and risks, the Active User perspective has drawn on methods from the humanities and certain parts of the social sciences, with a preference for qualitative methods such as interviews and field observations. This highlights the attitude that players are competent and selective, not just passive recipients of information. In other words, agency belongs to the player; the player, and not the medium, is in charge.

No two people experience a book, a film, a game, or any other medium in exactly the same way. The experience largely depends on their personal

background and the context in which they "absorb" the medium. An individual's experience with media can be completely unpredictable for researchers, no matter how thoroughly the media product itself has been analyzed and dissected. Thus, Active User researchers hesitate to draw a direct line linking a media product's content with a change in the behavior or attitude of the user.

Specifically, when someone plays a video game, the activity cannot be understood without considering the context he is playing in. The player may have friends hanging around, be alone in a big house, or be just stopping by the local game café. It seems problematic, these researchers argue, to insist that certain games are dangerous, as the meaning of playing a game is not a static entity to be discovered by looking at the game. As Carsten Jessen states:

> Computer games primarily acquire their meaning and content through their concrete use in concrete situations. In this sense they are more a kind of tool for social relations than a means of communicating the messages one normally looks for in the media. This should not be seen as a claim that there is no content in computer games, or that this does not matter; it should be taken as an indication that we cannot interpret a content outside the concrete practice which also provides the framework of understanding.[54]

When an outsider views a video game, assumptions are the easiest thing to make. A well-meaning researcher—not to mention an anxious parent—sees a child playing a video game, gleefully shooting other children, and eventually being shot himself. The adults may be horrified. For the child, this is play. The player is not really killing other children but competing against the like-minded, exploring worlds of intrigue and adventure, and socializing with others through a fascinating medium. If this defense of video games seems hard to stomach, remember that violent, unspeakable acts are abundant in fairy-tales, which hardly raise an eyebrow.

Theories and Methods

The Active User perspective cannot be understood as a close-knit research community; rather it is a varied group of researchers with a shared mindset. The perspective has been strongest in the Nordic countries and the UK. Researchers tend to draw on a variety of theories and methods, and do not see them as mutually exclusive.

Literary Theories of Reader Response and Reception

"Reader response" theories originate in literary studies, and assign more interpretive agency to the reader (or user) than to the text itself. This approach began with work on theories of aesthetic response by Hans Robert Jauss and Wolfgang Iser in the 1970s. These authors proposed that readers create meaning by engaging with texts in a variety of ways. For example, as

we have seen, readers unconsciously "fill in the gaps" while reading; they are able to follow the plot even though some information is usually missing. Readers do a lot of mental work when reading, using their encyclopedic knowledge of the world and their ability to relate cause and effect to interpret texts.

Another interesting branch of reader response theory is represented by Stanley Fish, who makes the radical proposal that texts have no meaning other than the meaning they are assigned by interpretive communities. For example, if a reader believes that Shakespeare is the peak of all literature, his ardor will unavoidably affect his reading.

All these perspectives insist, although in different ways, on the active participation of readers. Reader response theories have been exported to other media such as television, partly to contest the common view of audiences as passive consumers. It is also easy to export this point of view to video games. Theories of the Active User perspective all take for granted that engaging with video games is a worthwhile activity, and that a game's meaning is built only with a player's active participation.

Play as Meta-Communication[55]

Gregory Bateson points out that humans are capable of communicating that actions should be interpreted based on their context, and that play is crucial for developing this ability. The concept of "frame" refers to the context we use for interpreting a given activity; as a result, the meaning of a given activity depends on the persons involved. When a group of children is play-fighting in a backyard, a bystander may see it as aggressive and dangerous behavior. But for the children it can just as well be play; should someone be hurt by accident, it would not be interpreted as aggression but as an unfortunate side effect of playing. It is not possible to interpret another person's behavior without taking into account his own interpretation and perspective.

The Children's Perspective[56]

Flemming Mouritsen[57] and Carsten Jessen[58] attempt to get closer to children's use of video games. They see children as capable of constructing a frame around their play and culture that is almost impenetrable by adults. According to some researchers of children's culture, it is almost impossible for adults to see beyond their own frame of reference and approach video games with an open mind. Still, many studies from the Active User perspective attempt to understand the meaning of video games by seeing these games through children's eyes. The difficulty of this approach is obvious, but Active User researchers believe that the only way to really understand children's use of video games is to be a part of children's game culture—by observing, participating, and thus opening up themselves as researchers to other interpretations.

Catalyst Model[59]

Ferguson has become one of the strongest critics of the Active Media Perspective, and through the Catalyst model shifts focus toward the individual's disposition for being influenced by violent video games. The Catalyst Model stresses that individuals have different predispositions that are more important predictors (whether they be biological or environmental) of adverse effects from violent games than the violent video games themselves. As such, you can say that he accepts that media can have malign effects but that it has to be seen in complex interplay of different variables—and not as one size fits all.

Though the above theories vary widely in their origins and uses, all prioritize the individual, internal perspective of each human being, and stress that the creation of meaning is central to any media experience, including video games. As opposed to the Active Media perspective's positivist approach, the scientific standpoint of Active User researchers is hermeneutic: the aim of research is to constantly revisit a problem and construct an increasingly valid interpretation. The goal is not so much to generalize the results to all people but rather to offer reflections, insights, and understanding of a specific phenomenon that will be increasingly convincing, although never conclusive.

Most researchers in the broad Active User group will draw on some of these theoretical perspectives, although their influence may be implicit. Fish's theory, for example, is often seen as too extreme, but Iser's foundational reader response work still has an impact though researchers may not explicitly mention him.

Important Studies

It should be noted that the number of Active User studies that directly examine video games and risks is quite limited. Compared to the hundreds of studies coming from the Active Media camp over the last generation, Active User researchers have only made limited forays into the topic. As mentioned at the beginning of this chapter, many game researchers shy away from this subject as they find it hard to generalize from a single media object. Video games are so diverse, and mean so many different things to different people, that from an Active User perspective it hardly makes sense to ask the broad question: "Are video games dangerous?" or even a more humble question such as: "Do violent video games lead to aggression?" For one 5-year-old, a video game may be conceived as dangerous, violent, and scary, while another child can perceive the same game completely differently at the same age. Instead, the goal is to create a multifaceted picture of video games, rather than limiting research by replicating existing stereotypes and assumptions. We can see the Active User perspective at work in a few key examples.

In 2000, Birgitte Holm Sørensen and Carsten Jessen[60] studied 31 children in Denmark, aged from 5 to 17 years. The children were interviewed in

small groups, and then the researchers observed about 20 of them playing video games. The goal was simple: to arrive at a picture of children's perception of playing video games, their ability to distinguish between fiction and reality, and their appreciation of interactivity. The study found that the potential negative influence from violent content in video games is smaller than that of violence in other media. The researchers also see video games as close to ordinary play activities, and concluded that the children used video games as yet another toy.

Several other studies have turned their focus on adults, and found that adults often have a stereotyped and limited knowledge of video games. They therefore cannot predict how a person will understand a specific media product. They have little experience with video games and often approach the medium with a wish to censor, regulate, and change the experience into something they know and understand. From this perspective, computer games will seldom fulfill an adult's conception of an acceptable activity for children. As we saw in Chapter 3, adults have a tendency to idealize children's play; adults understand video games in relation to how they think children should play and behave.

An adult's interpretation may thus be very far from how children themselves understand the activity of playing video games.[61] Active User researchers insist that violent video game content cannot be taken at face value. Video games may on the surface look like simple, bloody fighting games, when in fact their basic story lines help us handle the difficult realities of life. In a hostile and scary world, video games may be just what a lot of children need to address their fears and insecurities.[62]

In the large European research project "Children, Young People and the Changing Media Environment," organized by Sonia Livingstone,[63] video games play a central part. The study looks at 12 European countries, and combines qualitative and quantitative methods. In total, close to 15,000 children aged between 6 and 16 participated. The researchers found that media increasingly plays a central role in private homes, and that video games are prominent in this.

Another important contribution to the field is the American psychologist Sherry Turkle's work, presented in her books *The Second Self*[64] and *Life on the Screen*.[65] Turkle has examined different perceptions of technology and tried to assess what computers and virtual worlds mean for us as human beings. She carried out a range of studies inspired by ethnography, and although the results are perhaps stretched too far, they are interesting. Turkle points out that computers inspire the user to think nonlinearly, and how the computer's interactive capacity forces the user to reconsider the relation between nothing less than man and machine, and life and death. According to Turkle,[66] people think that technology is merely a useful tool. Certainly technology is often useful; however, we fail to appreciate the more subtle ways it influences our lives every single day. Technology, including games, offers children new ways of thinking and relating to each other. Turkle finds that technology can have an effect but that this is probably

neither predominantly positive nor negative. This effect does not exist in a vacuum but is a part of everyday interactions and formed by each child and their surroundings.

The ambiguity of media is also attested to in another study, by Laura Ermi and Frans Mäyrä.[67] They maintain that video games can captivate children through features such as challenges, vast virtual worlds, and freedom. It may be this very freedom and the chance to get away from the restrictions of the real world that lead children to prefer video games to other play activities. This preference is a conscious one, as the researchers found that even their 10- to 12-year-old subjects could engage in a debate concerning the adverse effects of video games:

> In conclusion, the image of a child in contemporary game culture that emerges from our research is not one of helpless victim. On the contrary, many children seem to be very articulate about their preferences and capable of sharply criticizing games.[68]

Based on a recent study by Cheryl Olson, it seems that the Active User approach may be gaining ground in the United States, which has traditionally been more inclined toward the Active Media approach. Her focus is on describing the playing of video games through both qualitative and quantitative studies, and in relation to developmental stage, environment, and personality. She concludes: "Compared with other media such as books, films, and radio, electronic games appear to have an unusually expansive appeal and serve a surprising number of emotional, social, and intellectual needs."[69]

Games allow social interaction, joyful competition, peer learning, making friends, expressing creativity, and experimenting with different identities, and they serve as a means of relaxation or an outlet for difficult emotions. Based on the data collected, she argues that violent video games reflect a long tradition within toys more generally, and as such lie within the range of normal development. Violent video games are often favored not because of violent content but rather because they fit well with, in particular, boys' preferences for competition and mastery.

The above examples make clear that risk is far from central to the Active User perspective. However, the field's research does have some bearing on the relationship between video games and risks. Video games are used very differently and it is too simple to talk of general media effects.

Criticism

It seems that there is a double standard at work. When Active User researchers criticize the effect perspective, they challenge the reliability and validity of the research results. This is leveled at all areas of Active Media research, from its basic theoretical assumptions about how humans work, to the concrete ways of measuring effect. These rigorous demands are, however, rarely

applied to Active User researchers' own studies. This leads the Active Media perspective to adopt a dismissive attitude toward Active User studies. The former considers the experimental designs of the latter as fuzzy and lacking in scientific rigor; some Active Media proponents consider Active User work a lesser form of research (or, worse yet, not research at all).

More concretely, criticism of the Active User perspective is aimed at the limited scope of results and a weakness in documentation. This criticism, however, is usually not explicitly brought forward by the competing perspective. The Active Media perspective seems to have a limited interest in Active User studies, and finds that attacking the findings is not really worth much time and effort. As described earlier, Active User researchers, on the other hand, are quite ready to deliver criticism in the other direction.

It is clear that the theoretical framework outlined above makes it very hard (or even undesirable) for Active User researchers to generalize from their results. These researchers believe that observations of a group of children's reactions toward violence do not necessarily apply to other children. Still, data is sometimes stretched to provide general conclusions. This is problematic when you use qualitative research methods that are not capable of being—nor intended to be—used to support broader conclusions. A researcher may have performed observations in an after-school program over a couple of weeks, and formed an opinion on the children's interactions. These results cannot, however, be easily generalized as we have no way of knowing if these students are representative. The samples in Active User studies are not random, and are almost always too small to generalize to the whole population. Further, it is seldom possible for others to gain access to the exact data that form the researcher's opinions, and this makes it hard to test the conclusions of a given study.

OTHER QUESTIONS

The Active Media perspective has primarily examined the effect of violent video games on aggression in children and adolescents. However, other topics have been examined in relation to risks associated with video games and these are presented in the sections that follow.

The Content of Video Games

More and more people—academics and pundits alike—have questioned the unbalanced content of video games. The most pressing question is whether games present a stereotyped, perhaps even discriminatory, picture of the world.

The models for solving problems, stereotyped game universes, and frequent depictions of violence are just a few examples of problematic content. The Active Media studies examining this question offer various conclusions: they find that aggression is often a part of games, although the degree of violence depicted varies significantly both between video games and according

to the individual researchers analyzing content. Karen Dill and colleagues[70] have found that 79 to 85 percent of all video games contain violence, but Schierbeck and Carstens[71] stress that only 5 percent contain extreme violence. It makes a difference whether we are talking about the "war game," chess, or the almost vicious beat-'em-up *Mortal Kombat*.

Social Relations, Gender, and Exposed Groups

Another currently popular question is whether some groups are more exposed to the effects of video games than others. From the start, video games have appealed more to males than to females, and there are clear gender differences in video game preferences. As we saw in Chapter 4, the lack of strong female game characters has been lamented until (and also since) Lara Croft entered the scene. The difference gender makes has been closely investigated, especially in Justine Cassel and Henry Jenkins's *From Barbie to Mortal Kombat*, published in 1998. Recently, some researchers have bemoaned the lack of development in gender studies related to video games (see Chapter 6).

Studies on gender differences from the Active Media perspective have conflicting results. Some studies[72] find that females become more aggressive playing violent video games than males, while others do not find this effect.[73] The greater response of some women might be a result of female test subjects being less used to video games in general. They may therefore find it more intense, alien, and overwhelming to play a violent video game, which could trigger a more aggressive reaction.

Researchers from both the Active Media and Active User perspectives have stressed that age may play an important role in how someone is influenced by video games. Developmental psychology argues that young children (below 8 years) can have trouble distinguishing between reality and fiction. The lack of this ability is especially problematic if we look at children from the social learning theoretical perspective: children may imitate in real life what they see in video games. At least one Active User study, however, stresses that children may be capable early on of telling fiction from reality.[74] Dill and Dill[75] have challenged this claim, stating that the case for the existence of young, media-competent children may be exaggerated.

Extending this argument, Active Media researchers have claimed that video games affect our perception of the surrounding world. The results are remarkably in agreement, and show that violent video games result in a more negative perception of the real world.[76] A consequence of this may be that players perceive the world as a more hostile and dangerous place. They may also transfer behaviors from the game to the real world, leading to what can be labeled as crime.

It has been suggested that increases in crime may be a consequence of video games, although most studies supporting a link between crime and video games are based on arcade games (and violent crime in the United States has been decreasing for some time). Arcade games cost money to play,

and could therefore give players an incentive for finding what we might call alternative funding sources. This is, however, entirely different from most current video game playing, where one can play without incurring great cost. Some studies have also found that academic achievement can be affected by video games, but others have failed to find the same relationship.[77]

On the positive side, some have pointed out that players with low self-esteem may improve their sense of self by mastering video games. The relevant studies are now relatively old and have contradictory results. The conclusion—not surprisingly—is often that the relationship between self-esteem and video games varies depending on the player.[78] These correlations, however, have only been explored to a limited extent to date.

Addiction

Addiction to video games as a research topic has grown over the years since British psychologist Mark Griffiths and his co-researchers pioneered the field almost 20 years ago. Still, the definition of addiction causes a number of problems for these Active Media studies, and a satisfactory description has yet to be found. In the DSM-5, video game addiction was included under the classification Internet Gaming Disorder (but on the watch list warranting more research before official inclusion), and the methods for measuring this condition remain elusive. Indeed, a meta-study from 2013 on available protocols by King et al. found large inconsistences across 63 different studies. Video game addiction was measured with 18 different methods, and although the meta-study found that these methods could in the future form the foundation for a sound method, this is not the case yet. Rather, in these studies we see, for example, variations in coverage of key indicators for addiction, varying cutoff points for the appropriate clinical status, and lack of inter-rater reliability. In common words, the methods do not cover all areas of addiction and are not consistent between assessments by different testers, and therefore may result in different clinical conclusions.

Addiction to video games does not have the same repercussions as other forms of addiction. The secondary consequences that are normally associated with addiction, such as crime and debt, are not part of the pathology of video game addiction. Thus, labeling a person as a video game addict often seems to be based more on subjective values than scientific rationale. A reasonable conclusion on addiction is reached by Mark Griffiths:

> It is this author's belief that videogame addiction does indeed exist but that it affects only a very small minority of players. There appear to be many people who use video games excessively, but are not addicted as measured by these (or any other) criteria.[79]

Different definitions of addiction result in quite different estimates of the number of addicted players. In one study, 19.9 percent of players are found to display addictive behavior in relation to video games,[80] while in

another—by the same researcher, in fact—arrives at the significantly higher figure of 37.5 percent.[81] However, both of these numbers are adjusted downwards by the researchers, as most addicted players do not seem troubled by their playing pattern—nor does gameplay seem to harm their lives or their relationships. In a later article from 2005, Griffiths and Davies settle on the estimate that 7 percent of the total number of video game players are addicted, based on a definition of addiction as playing more than 30 hours a week, following the guidelines set out in the DSM-III-R protocol; as previously mentioned, however, these protocols have changed in the versions since the DSM-III-R.

The validity of Griffiths' early work was challenged in a study by Gentile in 2011.[82] In a survey, 1,178 American youths aged 8 to 18 were asked questions about game preferences. It was concluded that about 8.5 percent of video game players in this sample exhibited pathological patterns of play, as defined by a score based on the DSM-IV pathological gambling criteria. The study also found a correlation between young people exhibiting pathological gaming and attention deficit disorder. Indeed, pathological gamers were twice as likely to suffer from attention disorder, although one can only speculate about any causal relationship.

A later study, conducted in Singapore in 2011 with 3,034 schoolchildren, found a similar prevalence, of around 9 percent.[83] This was a two-year longitudinal study, which offered a unique opportunity to address the question of causality. The study found that greater amounts of gaming, lower social competence, and greater impulsivity seem to act as risk factors for becoming pathological gamers. It also found that the outcomes of pathological gaming were depression, anxiety, social phobias, and lower school performance. The study suggested that the relationship between pathological gaming and psychological disorders may be reciprocal, that is, they reinforce each other. Other studies have found similar findings, for example in the recent study by Brunborg, Mentzoni, and Frøyland (2014).

With the increasing number of studies on game addiction, Dara Kush and Mark Griffiths in 2012[84] decided to carry out a meta-study based on 30 studies of adolescents. The meta-study indicated that gaming addiction does exist despite what diagnosis method you use, and it can be categorized as a behavioral addiction rather than a disorder related to impulse control, which has implications for developing proper treatment. A meta-study from 2011 puts the most likely prevalence of gaming addiction to around 3.1 percent, which is considerably lower than most previous meta-studies. However, it was lent credence in another recent study with 8,807 European students, in which the prevalence of gaming addiction was noted at 3.62 percent (Strittmatter et al., 2015).

FINAL REMARKS

The potential adverse effects of video games have long been the subject of debate. This debate intensified in the 1990s as certain games became

still more realistic and detailed in their depictions of violence. In recent years, a number of violent incidents have been linked—especially in public debate—to the perpetrators' fascination with violent games. At the same time, more realistic audio and graphics, as well as more complex games, have reactivated worries about whether video games could damage those who play them.

We have summarized the volumes of research into the risks associated with video games, focusing specifically on the keystone of this research over the past two decades: the relationship between violent video games and increased aggression in gamers. The results have, however, often pointed in different directions. Whereas some researchers have found clear signs of aggressive behavior caused by video games, others have not been able to replicate their results, and still others have been strongly critical of the methods and underlying theories used in these studies.

This tangle of conflicting conclusions can be maddening, especially for parents desperate to protect their children, and governments eager to defend against any threats to a country's youth. But as we have seen, by dividing the research into two ideological camps we can gain a much better understanding of the nuts and bolts of this disagreement.

> The *Active Media* perspective is based on medical and psychological traditions. The researchers usually try to study the question with the help of specially staged experiments, in which people are exposed to different types of games under controlled (but not very realistic) conditions. The perspective has the strong support of certain academic groups in the US.
>
> The *Active User* perspective derives inspiration from ethnography and culturally oriented media studies. Researchers in this paradigm stress the fact that the user constructs meaning from each piece of media, and the process is always dependent on the context. According to this perspective, it is not possible to study the potentially adverse effect of violent video games in artificial settings, or without attempting to understand the player's own perspective. The Active User perspective is strongly represented in Scandinavia and Great Britain.

These two academic traditions are in fundamental disagreement over how (and even whether) to answer the question: "Do video games cause players any harm?" The majority of Active User researchers answer either that video games do not seem to cause harm in any direct sense, or, more likely, that the question is quite simply too general to answer. Some Active Media researchers, by contrast, answer in the affirmative, while others say that there might be a connection between video games and harm but are uncertain as to the extent.

We may never be able to provide a definitive answer, to this and to many other knotty problems. But after more than four decades of video games, decades of academic research, and untold hours of gameplay, we do know a few things. Video games are here to stay. Video games have been successful

because they are beguiling. They inspire the full gamut of emotional reactions, from fear and hatred to utter devotion. As they become more and more a part of our lives, and as they grow in directions we can today only barely imagine, the field of video game studies will need new talent, both to develop these creations and to seek to understand them. We hope that this book has given you a feel for the key avenues to understanding video games. And if you are intrigued, we invite you to explore further this astounding medium.



DISCUSSION QUESTIONS

- Which research perspective is best supported? Why is it difficult for researchers to engage in a fruitful debate?
- How would you criticize the Active User perspective?
- How would you criticize the Active Media perspective?
- What, if anything, makes the study of video games more complex than studying other media?
- Why is it important to research the malign effects of video games? What is your position on the issue? How would you research the potential malign effects?



RECOMMENDED GAMES

Mortal Kombat—one of the games to spark most controversy with its fatality moves, leading to studies on the risks associated with playing video games.

Doom—A very successful game to spark controversy at the time due its detailed graphics and violent content.

Grand Theft Auto V—The latest versions of the hugely successful game, where you play in an open-ended world, and can be just the bad boy you dream of.



FURTHER READINGS

Burgess, M., Dill-Shackleford, K., Stermer, P., Brown, B., & Brown, P. (2011, September). Playing with prejudice: The prevalence and consequences of racial stereotypes in video games. *Media Psychology*, *14*(3), 289–311.

(Estimated reading time: 40 minutes)

NOTES

1. Jenkins, 2007.
2. For an overview, see GamePolitics.com at http://gamepolitics.com/legislation.htm (accessed February 17, 2012).
3. Anderson, 2004, p. 113.
4. Heins et al., 2001.
5. Williams, 2003b.
6. Anderson, 2003.
7. Buckingham, 2001, p. 67.
8. Dill and Dill, 1998.
9. Calvert, 1999.
10. Barlett and Anderson, 2012.
11. Feshbach and Singer, 1971.
12. Bushman, Baumeister, and Stack, 1999; Olson, 2010.
13. Chandler, 1995; Gerbner, 1973.
14. Bandura, 1977.
15. Zillmann, 1979.
16. Berkowitz, 1984.
17. Anderson and Bushman, 2001.
18. Durkin and Barber, 2002, p. 373.
19. Gentile and colleagues, 2004.
20. Funk and colleagues, 2004.
21. Swing, Gentile, Anderson, and Walsh, 2010.
22. Williams and Skoric, 2005.
23. Anderson et al., 2008.
24. Kühn et al., 2018.
25. Przybylski and Weinstein, 2019.
26. Ferguson et al., 2017.
27. Cunningham, Engelstätter, and Ward, 2016.
28. Bensley and Van Eenwyk, 2001; Emes, 1997; Freedman, 2001; Robinson, Wilde, Navracruz, Haydel, and Varady, 2001.
29. Anderson and Bushman, 2001, p. 353.
30. Sherry, 2001.
31. Griffiths, 1999.
32. Freedman, 2001, p. 2.
33. Ferguson, 2009.
34. Ferguson, 2010a, p. 73.
35. Craig et al., 2010, p. 171.
36. Bensley and Van Eenwyk, 2001; Dill and Dill, 1998; Freedman, 2001; Goldstein, 2001; Ivory, 2001; Sherry, 2001; Ferguson, 2009.
37. Anderson et al., 2004; Anderson and Bushman, 2001.
38. Prescott, Sargent, and Hull, 2018.
39. For a summary, see Grossman and DeGaetano, 1999.
40. American Psychiatric Association, 2007.
41. American Psychiatric Association, 2015a.
42. American Psychiatric Association, 2015b.
43. American Psychiatric Association, 2015c.
44. http://henryjenkins.org/2007/04/a_few_thoughts_on_media_violen.html#sthash.WV5g1L4p.dpuf.
45. Durkin and Barber, 2002.
46. Anderson and Dill, 2000.
47. Ferguson, 2010a.
48. Geen, 2001.
49. 2001, p. 3.
50. Anderson and Dill, 2000.
51. Ferguson, 2007, p. 470.
52. Jenkins, 2000.
53. Gauntlett, 2001.
54. Jessen, 1998.
55. Bateson, 1972.
56. Jessen, 1998.
57. Mouritsen, 2003.
58. Jessen, 2001.
59. Elson and Ferguson, 2014.
60. Jessen, 2001; Sørensen and Jessen, 2000.
61. Jessen, 1998.
62. Jones, 2002.
63. Livingstone, 2002.
64. Turkle, 1984.
65. Turkle, 1995.
66. Turkle, 1995.
67. Ermi and Mäyrä, 2003.
68. Ermi and Mäyrä, 2003, p. 244.
69. Olson, 2010, p. 185.
70. Dill and colleagues, 2005.
71. Schierbeck and Carstens, 1999.
72. For example, Cooper and Mackie, 1986.
73. For example, Durkin and Barber, 2002.
74. Buckingham, 2000.
75. Dill and Dill, 1998.
76. Anderson and Dill, 2000; Kirsh, 2003; Lynch, Gentile, Olson, and Brederode, 2001.

77. Durkin and Barber, 2002; Harris, 2001.
78. Harris, 2001.
79. Griffiths and Davies, 2005.
80. Griffiths and Hunt, 1998.
81. Griffiths, 1997.
82. Gentile, 2009.
83. Gentile et al., 2011.
84. Kuss and Griffiths, 2012.